

BHAGYA SREE BOKKA

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SUMMARY

Senior Data Engineer with 5+ years of experience, skilled in Data Engineering, Analytics, and Database Administration, and committed to driving impactful solutions within distributed environments. Achieved a 75% reduction in data discrepancies by leveraging expertise in schema modeling, database design, and analysis. Developed and maintained Data Warehouse solutions across on-premise and cloud environments, ensuring robust enterprise-supported databases and software implementations. Spearheaded collaboration with design, development, and testing teams to drive projects end-to-end, accelerating timelines by 15% and implementing best practices to set teams up for success.

TECHNICAL SKILLS

Languages:	Python, C#, HTML, SQL, T-SQL, PL/SQL, VB.Net, PySpark, Matlab
Database Servers:	SQL Server, Snowflake, Oracle
ELT/ETL:	SSIS, Azure Data Factory, Informatica Power Center, AWS Glue, AWS Redshift, AWS Lambda
Analysis & Reporting Tools:	SSAS, SSRS, Tableau, Microstrategy, Power BI, Azure Synapse
Cloud Data Warehouse:	Snowflake, Azure Databricks
Scheduling & Workflow tools:	SQL Server Agent, Apache Airflow
Tools/Methodologies:	SQL Server Data Tools, Microsoft Visual Studio 2013/2015, SQL SERVER Management Studio, SQL Profiler, Tuning Advisor, MathWorks

PROFESSIONAL EXPERIENCE

UTARI, Fort Worth, Texas

Data Engineer

January 2024 - Present

- Developing a **MatLab** code that will automatically download healthcare data from OneDrive and webpage and format the data that can be used for analytical purposes.
- Responsible for developing **Matlab** code that will perform statistical analysis on the output by following the best software engineering practices.
- Pioneered two **research** projects on damage and life prediction of heterogeneous material systems, establishing and monitoring **research protocols**, conducting field tests, and collecting data.
- Developed and integrated large-scale, distributed machine learning system life cycles utilizing cutting edge open source technologies.
- Revamped the **Python** codebase to enhance data processing for micro-tensile testing devices, resulting in a 40% reduction in simulation time and increased accuracy for composite material analysis in Abaqus **GUI Scripting**.
- Skilled in updating **Python** code for the existing **machine learning algorithms** across various versions, ensuring compatibility and optimizing performance for seamless transitions and enhanced functionality.
- Acquired knowledge on **ROS** (Robot Operating System) and **Robot networking**. Implemented **ROS** launch files to run Python scripts which records **ROS** bag files for image, state, front & rear laser, point cloud and audio data streams.
- Collaborated with multi-disciplinary product development teams to find possibilities for performance improvement and incorporate trained models.

GGK Technologies (Innova group)

Senior Software Engineer

April 2022 - August 2023

- Engineered a **SQL query optimization project** that accelerated report generation speed by 25%, significantly benefiting financial analysis.
- Collaborated with 30+ IT project managers, database programmers, and multimedia programmers to maintain the company database's **integrity** and **security**.
- Developed clear, detailed reports and presentations on data analysis findings and project outcomes to support decision-making among stakeholders.
- Implemented a robust ETL framework using **SQL Server Integration Services (SSIS)**, processing complex healthcare claims data; improved data handling efficiency by 40% and facilitated seamless reporting in MS SQL 2012 databases.
- Intervened in multiple Transformation and data cleansing activities using Control flow and data flow tasks in **SSIS** packages during data migration. Pioneered 100+ SSIS Packages using **FUZZY LOOKUP** and Merge Table tasks.
- Used multiple types of Control Flow tools and Data Flow tools such as Container, Aggregate, Derived column and Conditional

Splitting according to Data scrubbing.

- Led strategic planning efforts for data engineering projects, defining timelines, setting milestones, and allocating resources to ensure on-time delivery and alignment with business goals.
- Applied Import and Export Wizard, Package Installation, and Package Designer in **SSIS**.
- Developed and maintained **VB scripts in SSIS** to handle complex data transformations, error handling, and automation, reducing manual intervention and streamlining ETL processes.
- Enhanced ETL workflows by embedding custom **VB scripts** to implement dynamic data processing rules, increasing ETL efficiency by **15%**.
- Worked on client reports and custom reports (On demand, Ad Hoc Reports, Frequency reports, Summary Reports, Sub Reports, Drill-down, Dynamic grouping, Cross-Tab, graphical) using SQL Server Reporting Services 2008R2/2008.
- Scheduled reports for Daily, Weekly, Monthly reports for different categories and regions based on business needs. Collaborated with Information Security Engineering teams to integrate security best practices into data processing pipelines, enhancing data confidentiality and integrity.
- Mentored and took up 30+ sessions to help new hires learn **SQL, Tableau, and Data Warehousing** concepts and develop functional and technical requirements.
- Led **data quality** initiatives by establishing data governance standards, performing data profiling, and implementing validation rules, improving data accuracy for decision-making by **40%**.
- Designed **data quality** frameworks to identify and remediate data issues in ETL processes, ensuring data consistency across systems and improving reporting reliability.

Software Engineer

January 2020 - March 2022

- Designed and developed a highly scalable data model and data warehouse using **Snowflake**, resulting in a 40% improvement in data processing speed and a 25% reduction in storage costs.
- Led design of a new **Snowflake**-based data warehouse, resulting in 40% improvement in data retrieval times and supporting 500+ users.
- Created **Snowpipe** for continuous data load and redesigned the views in **Snowflake** to increase the performance.
- Developed **Snowflake** procedures for executing branching and looping. Used COPY, LIST, PUT and GET commands for validating the internal stage files, Used import and Export from the internal stage (**Snowflake**) from the external stage.
- Automated 40+ processes to load files from 10 vendors, generate extracts in **SSIS**, and deployed **SSIS** Packages to **SQL server**, leading to optimized system performance and reduced query execution time by 25%.
- Configured 40+ workflows using **Apache Airflow** for the S3 bucket and **Snowflake** data warehouse and created **DAGS** to run the **Airflow**, causing a 30% reduction in data processing time and improved data accuracy.
- Designed and developed **ETL** integration patterns using **Python** on Spark, developed framework for converting existing PowerCenter mappings and to **PySpark**(Python and Spark) jobs. Provided guidance to the development team working on **PySpark** as ETL platform.
- Coordinated with business users by involving in design meetings and understanding & analyzing Business requirements in the **HealthCare domain**, which contributed to a 15% decrease in project timeline and efficient project workflow.
- Expanded team collaboration through activities such as code reviews, merging requests, addressing potential bottlenecks, and assisting whenever necessary, leading to a 20% drop in post-release bugs.
- Overcame complex data integration challenges to deliver high-quality, scalable database solutions aligned with business goals.
- Analyzed healthcare datasets to improve population health metrics, develop case management dashboards, and support Medicaid HEDIS reporting, contributing to a **15% increase in quality ratings**.
- Established and enforced data governance policies to ensure data quality, consistency, and compliance with industry standards.

Associate Software Engineer

April 2018 - December 2019

- Designed, coded, and implemented numerous **SQL Server** Databases, Tables, Indexes, Stored procedures, Views, Triggers in **SQL Server** using **T-SQL** with excellent troubleshooting, Query Optimization, SQL Profiler and resolution skills in **Microsoft SQL Server**.
- Created and established the Stored procedures, triggers to handle complex business rules, history data and audit analysis.
- Utilized advanced SQL techniques, including window functions, CTEs, and indexing, to optimize query performance and minimize resource usage in large data environments.
- Worked on multiple analysis tasks to improve **HEDIS** Ratings.
- Configured Mail Profile for sending automatic mails to the respective people when a job fails or succeeds. Maintained SQL Script for creation of Database Objects.
- Designed and optimized complex SQL queries to support data analysis, ETL processes, and reporting functions, improving data retrieval times and accuracy.
- Worked on automating the load processes to load the files received from multiple vendors. As part of this, developed numerous processes in **SSIS** and deployed **SSIS** Packages to **SQL server**.
- Worked on migrating all the **SSIS** solutions and SQL Agent jobs from One server to the other. Managed variable parameters in ETL processes to handle complex data transformation requirements for multiple projects.

- Implemented 50+ operational reports using **SSRS**. Created time-based and even-based subscriptions to deliver the reports to file share and emails using **SSRS**.
- Implemented executive dashboards using **Tableau**, developed & modified interactive dashboards and created guided navigation links within the dashboards in **Tableau**. Accomplished all tasks related to advance analysis actions, calculations, parameters, background images, maps, trend lines and statistics in **Tableau**. Migrated 100+ **SSIS** solutions, **SQL Agent jobs**, **SSRS**, and **Tableau** Reports from one server to another, ensuring minimal downtime and optimal performance.
- Managed and optimized SQL Server jobs using AutoSys to automate complex scheduling and monitoring tasks, reducing manual intervention and enhancing data processing efficiency by 20%.
- Initiated Query clarifications, discussions & demos with clients and documented 100% of code workflows and functionalities accompanied by business needs, resulting in a 30% drop in turnaround time for addressing client inquiries.

Trainee Software Engineer (Intern)

December 2017 - March 2018

- Achieved proficient knowledge in developing a mobile application using **C#**, **Asp.Net**, and **RESTful Web API**, leading to a 15% improvement in data-driven decision-making.
- Partly involved in screen design and development with **HTML5**.
- Analyzed, designed, tested and implemented new tables in **SQL Server 2012** for multiple modules. As part of this task, new Stored Procedures and related objects were developed using **T-SQL**.
- Created new queries for the customers against tables for verification and signoff purposes. Maintained Master scripts for Database objects and implemented logs in stored procedures.
- Spearheaded the design and modification of 40+ stored procedures, views, and functions, and monitored 70 **SQL Agent Jobs**, following a 20% decline in response time to data requests.
- Overhauled 80+ **SQL Server & Oracle** tasks to RPA developers, resulting in a 55% improvement in debugging efficiency and a 15% reduction in system downtime due to quicker issue resolution.
- Conducted in-depth research in the healthcare domain for a Proof of Concept (POC), which showed a 30% improvement in automation process accuracy.

EDUCATION

University of Texas at Arlington, Arlington, Texas

Master of Science in Computer Science

CVR College of Engineering, Hyderabad, Telangana

Bachelor of Technology in Electronics and Communication Engineering

CERTIFICATIONS

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| • Hands-On Essentials: Data Warehousing Workshop - Snowflake (August 2024) | • Certified Neo4J Professional (June 2020) |
| • Microsoft Certified Azure Data Fundamentals (December 2021) | • Six Sigma Yellow Belt Professional by 6Sigmastudy (May 2020) |
| • Microsoft Certified Azure Fundamentals (November 2021) | • Project Management Essentials Certified (May 2020) |
| • Microstrategy Certified Analyst (June 2020) | • Scrum Fundamentals Certified by SCRUMstudy (May 2020) |
| | • Lean Six Sigma White Belt Certified (May 2020) |

ACHIEVEMENTS AND VOLUNTEERING WORK

- Awarded for **Merit** performance for academic excellence from the Bharat Dynamics Employees Educationally and Socially Backward Class Welfare Association in 2012 and CVR College of Engineering in 2015.
- Facilitated 20+ awareness campaigns with **Street Cause**, a student-run NGO, to educate and empower society to encompass a brighter tomorrow.
- Organized 10+ events as an **IETE** student coordinator at CVR College of Engineering from 2015 to 2018.
- Authored an article on “**Role of Woman Engineers in Technological Innovations for Nation Building**” in the **29th Indian Engineering Congress**, Woman Engineers Meet, organized by the **Institution of Engineers**, hosted by Andhra Pradesh State Centre in December 2014.
- Trained as a **Bala Vikas** student in **Sri Sathya Sai Seva Organization** in 2003, completed the **Group-II** examination in 2006, and promoted 70+ social service activities such as **Narayana Seva**.